FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Biology — Class IX | Paper: SSC-I

Syllabus: Chapter 7 — Metabolism

ANSWER KEY (Version 1)

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Q1. Sum of all chemical reactions in a living cell: **C — Metabolism** **1 mark**

Metabolism is the sum of all chemical reactions in each cell for maintenance of life. **Catabolism** and **anabolism** are the two types of metabolism, not metabolism itself. Enzymology is the study of enzymes.

Q2. Enzymes are produced by: **B — Living cells** **1 mark**

Enzymes are **biologically active globular proteins made by living cells**. Each cell synthesises its own enzymes, determined by active genes. They cannot be produced by dead cells or inorganic ions.

Q3. Part of enzyme where substrate binds: **C — Active site** **1 mark**

The **active site** is the charge-bearing cavity on the enzyme's surface where substrate binds and reaction occurs. The **allosteric site** is where non-competitive inhibitors bind — not the reactive site.

Q4. Enzyme loses function at high temperature because it is: **B — Denatured** **1 mark**

At very high temperatures, enzyme atoms vibrate vigorously — the globular structure and active site are destroyed (**denatured**). At low temperature, inactivity is reversible; denaturation at high temperature is permanent.

Q5. Competitive inhibitor binds to the: **C — Active site** **1 mark**

In **competitive inhibition**, the inhibitor resembles the substrate in shape and occupies the **active site**, blocking the substrate. The allosteric site is the target for non-competitive inhibitors only.

Q6. ATP is energy currency because it: **B — Stores and releases energy** **1 mark**

ATP is an **energy-rich compound** that stores energy (in high-energy phosphate bonds) and releases it on demand when hydrolysed — used like money across all cell functions.

Q7. Third component of ATP alongside adenine and ribose: **B — Three phosphate groups** **1 mark**

ATP = Adenosine + **Three phosphate groups** (triphosphate chain). The covalent bonds between phosphate groups are **high-energy bonds** that store the usable energy.

Q8. Oxygen is released as by-product through: **B — Photolysis of water** **1 mark**

In light-dependent reactions, water splits into O, H^+ and electrons — this is **photolysis of water**. Oxygen exits through stomata. Calvin cycle and Krebs cycle do not produce oxygen.

Q9. Calvin cycle takes place in the: **C — Stroma of chloroplast** **1 mark**

Light-independent (dark) reactions take place in the **stroma of chloroplast**. Light-dependent reactions are in thylakoid membranes. Glycolysis is in cytoplasm; Krebs cycle is in mitochondria.

Q10. Net ATP gain from glycolysis: **C — 2 ATP** **1 mark**

Glycolysis produces 4 ATP gross but consumes 2, giving a **net gain of 2 ATP** per glucose molecule. The 36 ATP figure is the total from complete aerobic respiration, not glycolysis alone.

Q11. Alcoholic fermentation (yeast) products: **C — Carbon dioxide** **1 mark**

In yeast: Pyruvic acid → **Ethyl Alcohol + Carbon dioxide**. This is used in wine making and bread baking. Lactic acid is the product in muscle cells, not in yeast.

Q12. Final electron acceptor in ETC: **C — Oxygen** **1 mark**

In the electron transport chain, **oxygen** is the final electron acceptor. It combines with H^+ ions to produce water. Without oxygen, the ETC stops — this is why aerobic respiration requires oxygen.

SECTION B — Short Questions Answers (3 marks each)

2(i) Metabolism / catabolism / anabolism with examples **3 marks**

Metabolism **1 mark**: Sum of all chemical reactions in each cell of living organisms for maintenance and sustainability of life.

Catabolism **1 mark**: Breaks down complex molecules into simpler ones, releasing energy. Example: Cellular respiration — glucose → CO_2 + H_2O + energy.

Anabolism **1 mark**: Builds complex molecules from simpler ones, using energy. Example: Photosynthesis — CO_2 + H_2O → glucose.

OR

Why enzymes are biological catalysts; two characteristics **3 marks**

Enzymes are **biological catalysts** **1 mark**: made by living cells; speed up reactions without being consumed or permanently changed.

Characteristic 1 **1 mark**: **Increase rate of reaction** — millions of times faster than non-enzymatic reactions; lower activation energy.

Characteristic 2 **1 mark**: **Required in small quantity** — enzyme structure regains original shape after reaction; one enzyme catalyses a huge amount of substrate repeatedly.

2(ii) Three types of cofactors with examples **3 marks**

Activators **1 mark**: Detachable inorganic ions. Example: Cl^- ions increase salivary amylase activity; Zn^{2+}, Fe^{2+}, Cu^{2+}.

Prosthetic groups **1 mark**: Organic molecules tightly and permanently attached to enzyme. Examples: FAD (Flavin Adenine Dinucleotide), Haem group.

Coenzymes **1 mark**: Detachable organic molecules. Examples: NAD (nicotinamide adenine dinucleotide), Coenzyme A, Vitamins A and C.

OR

Enzymes are specific — explanation with example **3 marks**

Specificity **1 mark**: Each enzyme catalyses only one type of reaction and will not act on a different substrate.

Reason **1 mark**: The **active site** has a unique shape fitting only one specific substrate — like a lock fits only one key.

Example **1 mark**: **Amylase** acts only on starch (not proteins or fats). **Pepsin** digests only proteins in the stomach.

2(iii) Mechanism of enzyme action: ES and EP complex **3 marks**

Step 1 **1 mark**: Specific substrate binds to the **active site** → forms **Enzyme-Substrate (ES) complex**.

Step 2 **1 mark**: Enzyme catalyses the reaction, converting substrate to product → forms **Enzyme-Product (EP) complex**.

Step 3 **1 mark**: EP complex breaks up; products released; **enzyme is unchanged** and available again for the next reaction.

OR

Energy of activation; how enzymes lower it **3 marks**

Activation energy **1 mark**: Minimum extra energy required to activate molecules to start a chemical reaction.

How enzymes lower it **1 mark**: By changing the shape and charge of the substrate at the active site, or providing an entirely different reaction mechanism.

Effect **1 mark**: Reactions can proceed at **body temperature** (37°C) without requiring extreme heat or harsh conditions.

2(iv) Effect of temperature on enzyme activity **3 marks**

Optimum temperature **1 mark**: Each enzyme works best at a specific optimum temperature. For human enzymes: 36–38°C. Rate increases with temperature up to this point.

Very high temperature **1 mark**: Enzyme atoms vibrate vigorously → globular structure and active site damaged → **denaturation** (permanent loss of activity).

Very low temperature **1 mark**: Insufficient activation energy → enzyme becomes **inactive** (reversible — activity resumes when temperature is restored).

OR

pH effect on enzyme activity; optimum pH for pepsin and trypsin **3 marks**

Optimum pH **1 mark**: Each enzyme performs best at specific pH. Change in pH affects ionisation of amino acids at the active site. Slight deviation reduces reaction rate; extreme pH denatures enzyme.

Pepsin **1 mark**: Optimum pH **2.00** (acidic). Works in stomach; pH maintained by HCl.

Trypsin **1 mark**: Optimum pH **8.00** (alkaline). Works in small intestine. Both are protein-digesting enzymes.

2(v) Competitive vs non-competitive inhibition 3 marks

Feature	Competitive	Non-competitive
Binding site	**Active site** 1 mark	**Allosteric site** 1 mark
Shape similarity to substrate	Yes — resembles substrate	No
Reversibility 1 mark	Reversible with excess substrate	Irreversible; active site deformed

OR

Allosteric site; how non-competitive inhibitor damages active site 3 marks

Allosteric site 1 mark: A point on the enzyme surface other than the active site where non-competitive inhibitors attach.

Mechanism 1 mark: Inhibitor binding alters the **globular shape** of the enzyme → deforms and damages the active site.

Result 1 mark: Substrate **cannot bind** to the damaged active site; no ES complex forms; reaction stops.

2(vi) Structure of ATP molecule; three components 3 marks

Adenine 1 mark: Double-ringed nitrogen base. Adenine + Ribose = Adenosine.

Ribose sugar 1 mark: Five-carbon (pentose) sugar.

Three phosphate groups 1 mark: (PO_4) linked in a triphosphate chain. Bonds between phosphate groups are **high-energy bonds** (wavy lines) — energy stored here is used by cells for all life processes.

OR

ATP–ADP cycle; energy storage and release 3 marks

Energy storage 1 mark: ADP + inorganic phosphate (P_i) combine by **condensation** to reform ATP. Energy from catabolism (or light in plants) drives this — stored in phosphate bond.

Energy release 1 mark: ATP is **hydrolysed** — third phosphate separates → releases energy + ADP + P_i.

Significance 1 mark: Energy available on demand for movement, active transport, growth, nerve impulse, gland secretion — without wasteful overproduction.

2(vii) Photosynthesis equation; raw materials and products 3 marks

$$6CO_2 + 12H_2O \xrightarrow{\text{Light/Chlorophyll}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

Raw materials 1 mark: CO_2 (from air via stomata) and H_2O (absorbed by roots). Chlorophyll captures light energy.

Products 1 mark: **Glucose** (used for energy and building materials) and **Oxygen** (by-product released through stomata).

Conditions 1 mark: **Light energy** and **chlorophyll** are essential. Photosynthesis occurs in plants, algae, and cyanobacteria.

OR

Three limiting factors of photosynthesis 3 marks

Light intensity 1 mark: More light intensity and longer duration increase photosynthesis rate by providing more energy for light reactions.

CO_2 concentration 1 mark: CO_2 is raw material (substrate) for the photosynthetic reaction. More CO_2 increases rate — but only up to a specific limit.

Temperature 1 mark: Optimum temperature for photosynthesis is 25°C. Temperature below or above the optimum decreases rate by affecting enzyme activity.

2(viii) Light-dependent reactions: location and products 3 marks

Definition 1 mark: Reactions that require light energy directly; they depend on Photosystems I and II to absorb light and begin photosynthesis.

Location 1 mark: **Thylakoid membranes** (grana) of chloroplasts, where photosynthetic pigments are arranged into photosystems.

Products 1 mark: **ATP**, **NADPH**, and **Oxygen** (from photolysis of water). ATP and NADPH power the Calvin cycle.

OR

Photolysis of water; its role in light-dependent reactions 3 marks

Photolysis 1 mark: Water molecule splits into oxygen atoms, $2H^+$ ions and 2 electrons under light energy absorbed by Photosystem II.

Role 1 1 mark: Electrons released by photolysis **replace electrons lost** from the excited chlorophyll of Photosystem II — maintaining the reaction.

Role 2 1 mark: Oxygen atoms are released as **O_2** through stomata (source of atmospheric oxygen); H^+ ions reduce $NADP^+$ to NADPH.

2(ix) Four stages of aerobic respiration in order 3 marks

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1. GLYCOLYSIS (Cytoplasm):  Glucose → 2 Pyruvic Acid + 2 ATP + 2 NADH
2. FORMATION OF ACETYL CoA (Mitochondria): 2 Pyruvic Acid → 2 Acetyl CoA + 2CO₂ + 2 NADH
3. KREBS CYCLE (Mitochondrial matrix): 2 Acetyl CoA → 4CO₂ + 2 ATP + 6 NADH + 2 FADH₂
4. ELECTRON TRANSPORT CHAIN (Inner mitochondrial membrane): NADH + FADH₂ → 32 ATP + H₂O
```

1 mark four stages named correctly; 1 mark locations; 1 mark products at each stage.

OR

Aerobic respiration equation; ATP yield per glucose 3 marks

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + \text{Energy (ATP)}$$

ATP yield 1 mark: Complete aerobic respiration yields **36 ATP** per glucose (2 glycolysis + 2 Krebs + 32 ETC).

Comparison 1 mark: Aerobic = 36 ATP vs anaerobic = 2 ATP — aerobic is 18 times more efficient.

Requirement 1 mark: Requires **oxygen** as final electron acceptor; without O_2, only glycolysis + fermentation occurs.

2(x) Glycolysis: definition, location, products 3 marks

Definition 1 mark: First stage of cellular respiration — one glucose (6C) broken down into two molecules of **pyruvic acid** (3C each).

Location 1 mark: **Cytoplasm** of the cell, outside mitochondria. Common to both aerobic and anaerobic respiration.

Products 1 mark: 2 pyruvic acid + **2 ATP** (net) + 2 NADH.

```
Glucose → 2 Pyruvic Acid + 2 ATP + 2 NADH
```

OR

Role of Krebs cycle; products formed 3 marks

Role 1 mark: Cyclic process in **mitochondrial matrix**; acetyl CoA is fully oxidised; energy extracted as electron carriers (NADH, $FADH_2$) that feed the ETC.

Products 1 mark: **$4CO_2$ + 2 ATP + 6 NADH + 2 $FADH_2$** per glucose.

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2 Acetyl CoA → 4CO₂ + 2 ATP + 6 NADH + 2 FADH₂
```

Significance 1 mark: NADH and $FADH_2$ carry electrons to ETC where most ATP is produced; CO_2 released as waste.

2(xi) Alcoholic vs lactic acid fermentation 3 marks

Feature	Alcoholic Fermentation	Lactic Acid Fermentation
Organism	Yeast, some bacteria	Muscle cells; lactic acid bacteria
Products 1 mark	Ethyl alcohol + CO_2	Lactic acid only
Equation 1 mark	Pyruvic acid → Ethyl Alcohol + CO_2	Pyruvic acid → Lactic Acid
Example 1 mark	Wine making; bread baking	Yogurt/cheese; muscle fatigue in exercise

OR

Four reasons anaerobic respiration is important 3 marks

1. 1 mark Early Earth had no free oxygen; earliest organisms produced energy by **anaerobic respiration** — an evolutionary necessity.
2. 1 mark Bacteria use anaerobic respiration for **cheese and yogurt making** — industrially important food production.
3. 1 mark **Yeast fermentation** used in wine making and baking industry (CO_2 causes bread to rise).
4. Lactic acid fermentation provides **emergency energy** to muscle cells during vigorous exercise when oxygen supply is insufficient.

SECTION C — Long Questions Answers (5 marks each)

3(i) Define enzyme; five characteristics 5 marks

Enzyme 1 mark: Biologically active globular proteins made by living cells which tremendously speed up biochemical reactions — the biological catalysts of cells. Study of enzymes = **enzymology**.

1. Globular proteins: Chains of amino acids folded into 3D globular proteins. Few are RNA — called **ribozymes**.

2. Increase rate of reaction 1 mark: Millions of times faster than non-enzymatic reactions. Only speed up; do not start reactions. They lower activation energy.

3. Required in small quantity 1 mark: Enzyme regains original shape after each reaction → reused repeatedly → small quantity catalyses a huge amount of substrate.

4. Specific 1 mark: Each enzyme catalyses only one reaction on one substrate. Active site has unique shape. Example: **amylase** acts only on starch.

5. Require cofactor 1 mark: Many enzymes need a non-protein cofactor — activators (Zn^{2+}, Cl^-), prosthetic groups (FAD, Haem), or coenzymes (NAD, Coenzyme A).

Also valid: energy of activation; metabolic pathways; regulation of production and activity.

OR

Three factors affecting enzyme activity (temperature, pH, substrate concentration) 5 marks

Temperature 1 mark: Rate rises with temperature to **optimum** (36–38°C for humans). Above optimum: enzyme atoms vibrate → globular structure damaged → **denaturation** (permanent). Below optimum: enzyme inactive (reversible); insufficient activation energy.

pH 1 mark: Each enzyme has specific **optimum pH**. Change affects ionisation of amino acids at active site. Pepsin: pH 2 (stomach); Trypsin: pH 8 (small intestine). Extreme pH → denaturation.

Substrate concentration 2 marks: Rate is directly proportional to substrate concentration when enzyme molecules are sufficient. Rate increases until **saturation of active sites** — all sites occupied. Beyond saturation, more substrate has no effect. This plateau is the maximum rate (Vmax).

Inhibitors & Activators 1 mark: **Inhibitors** (poisons, cyanide, antibiotics, drugs) reduce/stop enzyme activity. **Activators** increase rate. Both regulate enzymes from inside or outside the cell.

3(ii) Mechanism of enzyme action; activation energy; lock-and-key vs induced-fit 5 marks

Active site 1 mark: Charge-bearing cavity on enzyme surface — very small, specific shape. Substrate binds here and reaction occurs.

Mechanism — 3 steps 1 mark: (1) Substrate binds active site → **ES complex**. (2) Enzyme catalyses reaction → **EP complex**. (3) Products released → enzyme unchanged and reusable.

Lowering Activation Energy 1 mark: Enzyme changes shape/charge of substrate at active site or provides a different reaction mechanism → reactions proceed at body temperature without extreme conditions.

Lock-and-Key Model 1 mark: Emil Fischer. Active site is **rigid and fixed** (lock); substrate is the specific key. Explains why one enzyme fits only one substrate.

Induced-Fit Model 1 mark: More accepted. Active site is **flexible**. When the correct substrate approaches, active site **changes shape** to accommodate it precisely. The site is still specific — only the correct substrate can induce the correct conformational change.

OR

Competitive and non-competitive inhibition; allosteric site; daily-life examples 5 marks

Enzyme inhibition 1 mark: Chemicals at the reaction site prevent enzymes from performing their role. An inhibitor binds to the enzyme in place of the substrate but does not convert to products. External factors: poisons, cyanide, antibiotics, drugs, accumulated products.

Competitive inhibition 2 marks: Inhibitor's structure resembles substrate → occupies **active site** → substrate blocked → no product. **Reversible**: excess substrate displaces inhibitor. Example: Sulfonamide drugs competitively inhibit bacterial enzyme needed for folic acid synthesis → kills bacteria (antibiotic action).

Non-competitive inhibition 2 marks: Inhibitor does not resemble substrate → binds to **allosteric site** (enzyme surface, not active site) → alters globular shape → active site damaged → substrate cannot bind → reaction stops. **Irreversible**. Example: **Cyanide** (rat poison) inhibits cytochrome c oxidase in ETC. Heavy metals (Pb, Hg) can also act as non-competitive inhibitors.

3(iii) ATP as energy currency: structure, ATP–ADP cycle, uses in body 5 marks

ATP as Energy Currency 1 mark: Living organisms use and store energy at cellular level in the form of **Adenosine Triphosphate (ATP)**. Called energy currency because it stores/releases energy like money — converted between forms as needed. Human cells use 100–150 moles of ATP per day.

Structure of ATP 1 mark: Three components: (1) **Adenine** (double-ringed N-base) + (2) **Ribose sugar** (5C pentose) = Adenosine. (3) **Three phosphate groups** (PO_4) in triphosphate chain. Bonds between phosphate groups = **high-energy bonds** (wavy lines). AMP = 1 phosphate; ADP = 2 phosphates; ATP = 3 phosphates.

ATP–ADP Cycle 2 marks:

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Energy from catabolism / light (plants)        Energy for cellular work
         ↓                                               ↓
ADP + Pᵢ ──condensation──► ATP ──hydrolysis──► ADP + Pᵢ
(bond formed = energy stored)          (bond broken = energy released)
```

Uses of ATP 1 mark: Working of muscles, nerve impulse transmission, growth and repair of cells, active transport, gland secretion. **Brain** uses maximum ATP — 25% of total body ATP.

OR

Photosynthesis: equation, light-dependent reactions, Calvin cycle in sequence 5 marks

Overall equation 1 mark:

$$6CO_2 + 12H_2O \xrightarrow{\text{Light/Chlorophyll}} C_6H_{12}O_6 + 6H_2O + 6O_2$$

Takes place in plants, algae, cyanobacteria. Chlorophyll absorbs light; CO_2 and H_2O are raw materials; glucose and O_2 are products.

Light-dependent reactions (Phase 1) 2 marks: In **thylakoid membranes** (grana). Starts at **Photosystem II**: (1) Chlorophyll absorbs light → emits electron pair → accepted by primary electron acceptor. (2) **Photolysis of water**: $H_2O \rightarrow O_2 + 2H^+ + 2e^-$ (O_2 released through stomata). (3) Electrons pass through **ETC** → energy for **ATP synthesis**. (4) **Photosystem I**: electrons + $2H^+$ reduce $NADP^+$ → **NADPH**. Products: ATP, NADPH, O_2.

Calvin cycle / Light-independent reactions (Phase 2) 2 marks: In **stroma of chloroplast**. Uses ATP and NADPH from Phase 1. Steps: (1) CO_2 + RuBP (5C) → 6C intermediate → 2 × 3-PGA. (2) 3-PGA reduced by NADPH + ATP → G3P (3C sugar). (3) G3P → glucose synthesis + RuBP regeneration for next cycle. Described by **Melvin Calvin**.

3(iv) Aerobic vs anaerobic respiration; glycolysis; fate of pyruvic acid 5 marks

Feature 1 mark	Aerobic Respiration	Anaerobic Respiration
Oxygen needed	Yes	No
Glucose oxidation	Complete → CO_2 + H_2O	Incomplete → alcohol / lactic acid
ATP yield	36 ATP	2 ATP
Site	Cytoplasm + Mitochondria	Cytoplasm only

Glycolysis — Common first stage 1 mark: Both pathways start with **glycolysis** in cytoplasm. Glucose (6C) → 2 Pyruvic acid (3C) + 2 ATP + 2 NADH.

Fate in Anaerobic respiration 1 mark: Pyruvic acid not fully oxidised.

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Yeast:    Pyruvic Acid → Ethyl Alcohol + CO₂   (alcoholic fermentation – wine, bread)
Muscles: Pyruvic Acid → Lactic Acid             (lactic acid fermentation – yogurt, exercise fatigue)
```

Total ATP = only 2 (from glycolysis).

Fate in Aerobic respiration 2 marks: Pyruvic acid enters mitochondria → fully oxidised in 3 further stages:

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Stage 2: 2 Pyruvic Acid + 2 CoA → 2 Acetyl CoA + 2CO₂ + 2 NADH
Stage 3 (Krebs): 2 Acetyl CoA → 4CO₂ + 2 ATP + 6 NADH + 2 FADH₂
Stage 4 (ETC):   10 NADH + 2 FADH₂ → 32 ATP + H₂O  (O₂ = final e⁻ acceptor)
Total: 2 + 2 + 32 = 36 ATP per glucose molecule
```

OR

Aerobic respiration: four stages with ATP yield at each 5 marks

Stage 1 — Glycolysis 1 mark: Cytoplasm. Glucose → 2 Pyruvic acid. Net yield: **2 ATP + 2 NADH**. Common to aerobic and anaerobic.

Stage 2 — Formation of Acetyl CoA 1 mark: Mitochondria. Pyruvic acid oxidised; 2C acetyl group + CoA → Acetyl CoA; CO_2 released; NADH produced. Yield: **2 NADH** per glucose.

```
2 Pyruvic Acid + 2 CoA → 2 Acetyl CoA + 2 CO₂ + 2 NADH
```

Stage 3 — Krebs Cycle 1 mark: Mitochondrial matrix. Acetyl group fully oxidised. Yield: **4 CO_2 + 2 ATP + 6 NADH + 2 $FADH_2$**.

Stage 4 — Electron Transport Chain (ETC) 2 marks: Inner mitochondrial membrane. Electrons from NADH and $FADH_2$ pass through carrier molecules releasing energy → ATP synthesis. NADH → 3 ATP each; $FADH_2$ → 2 ATP each. Final electron acceptor = **oxygen** → combines with H^+ → **water**.

```
Glycolysis  =  2 ATP
Krebs cycle =  2 ATP
ETC         = 32 ATP  (10 NADH × 3 + 2 FADH₂ × 2, minus ETC costs)
────────────────────
TOTAL       = 36 ATP per glucose molecule
```